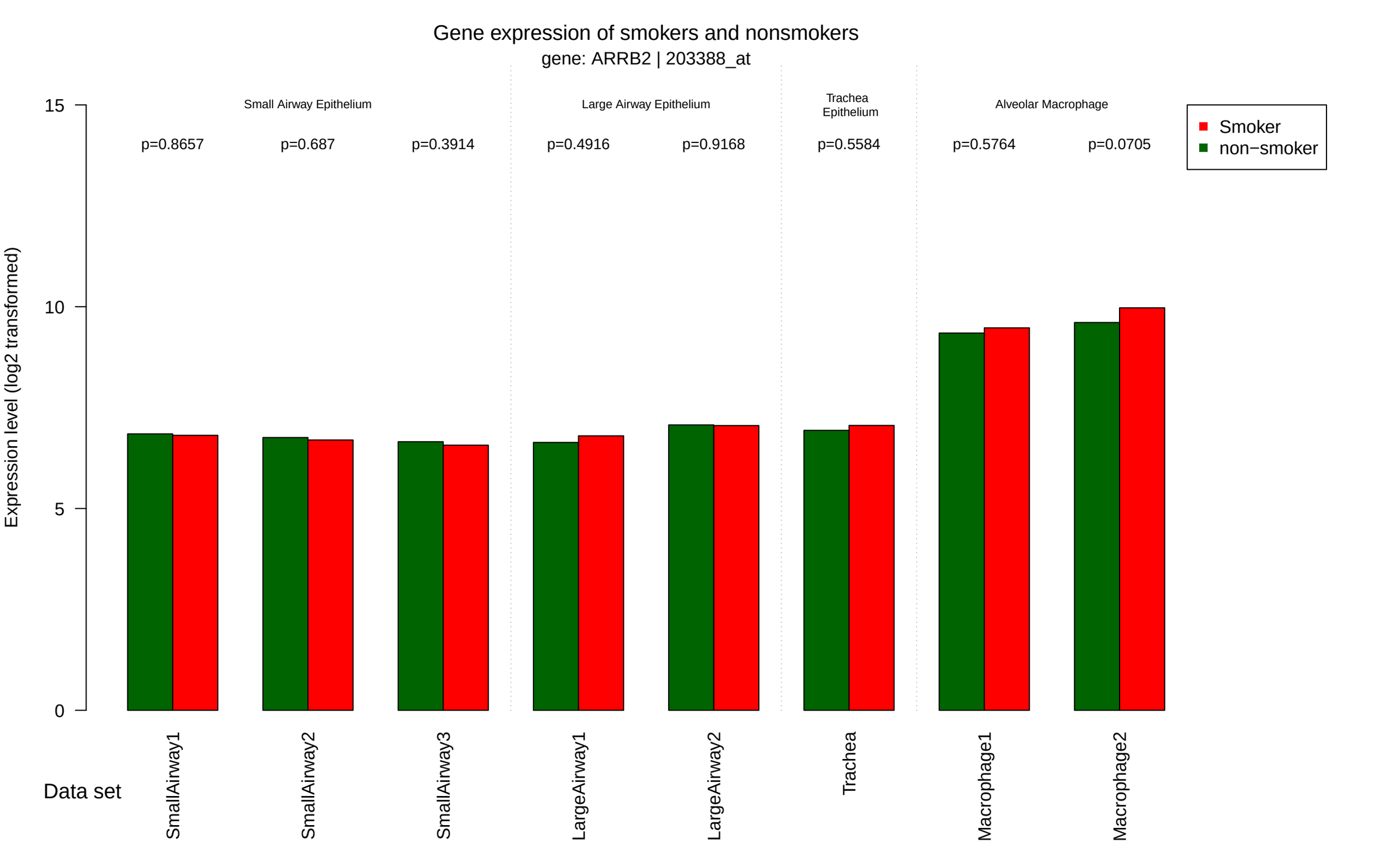
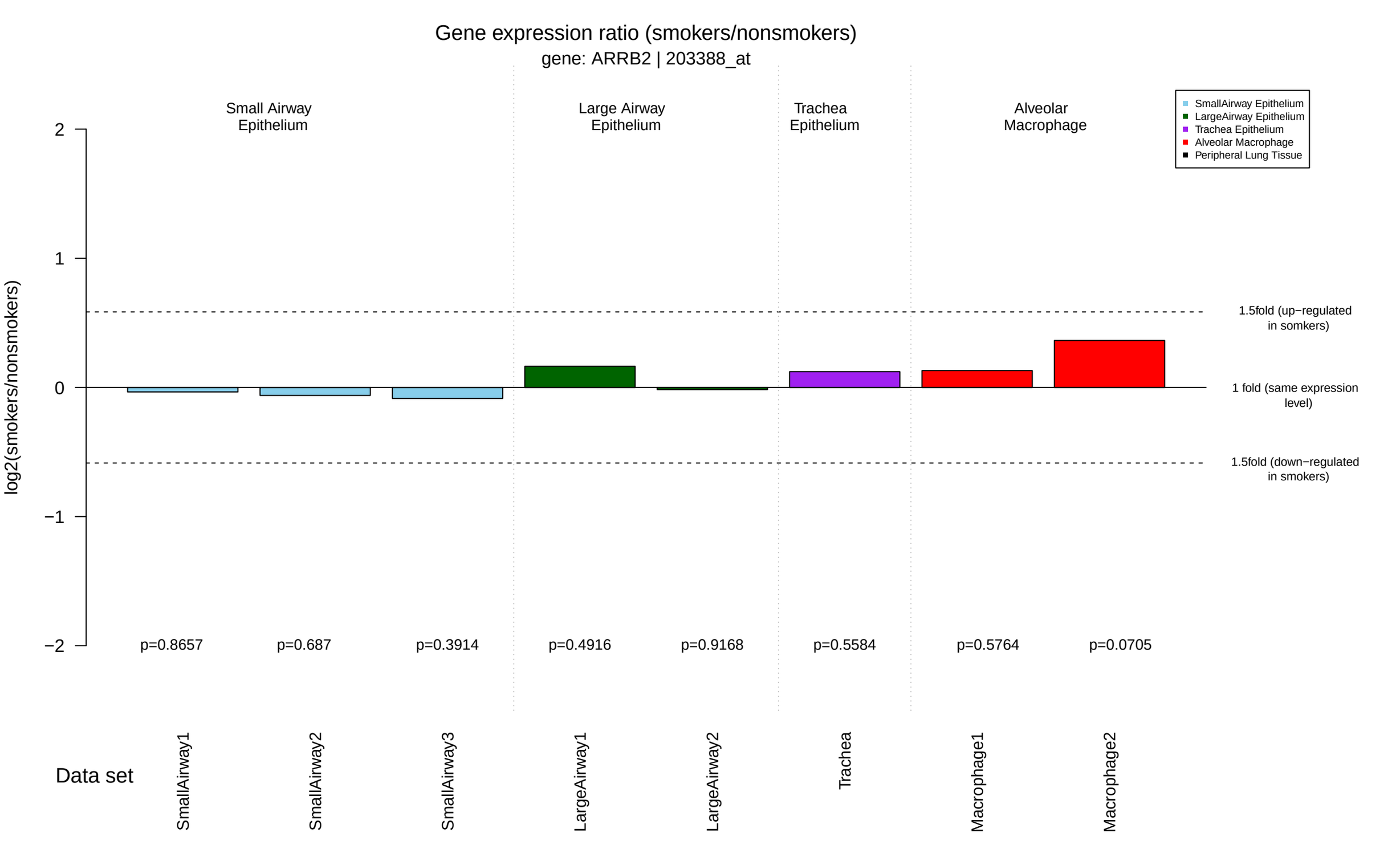
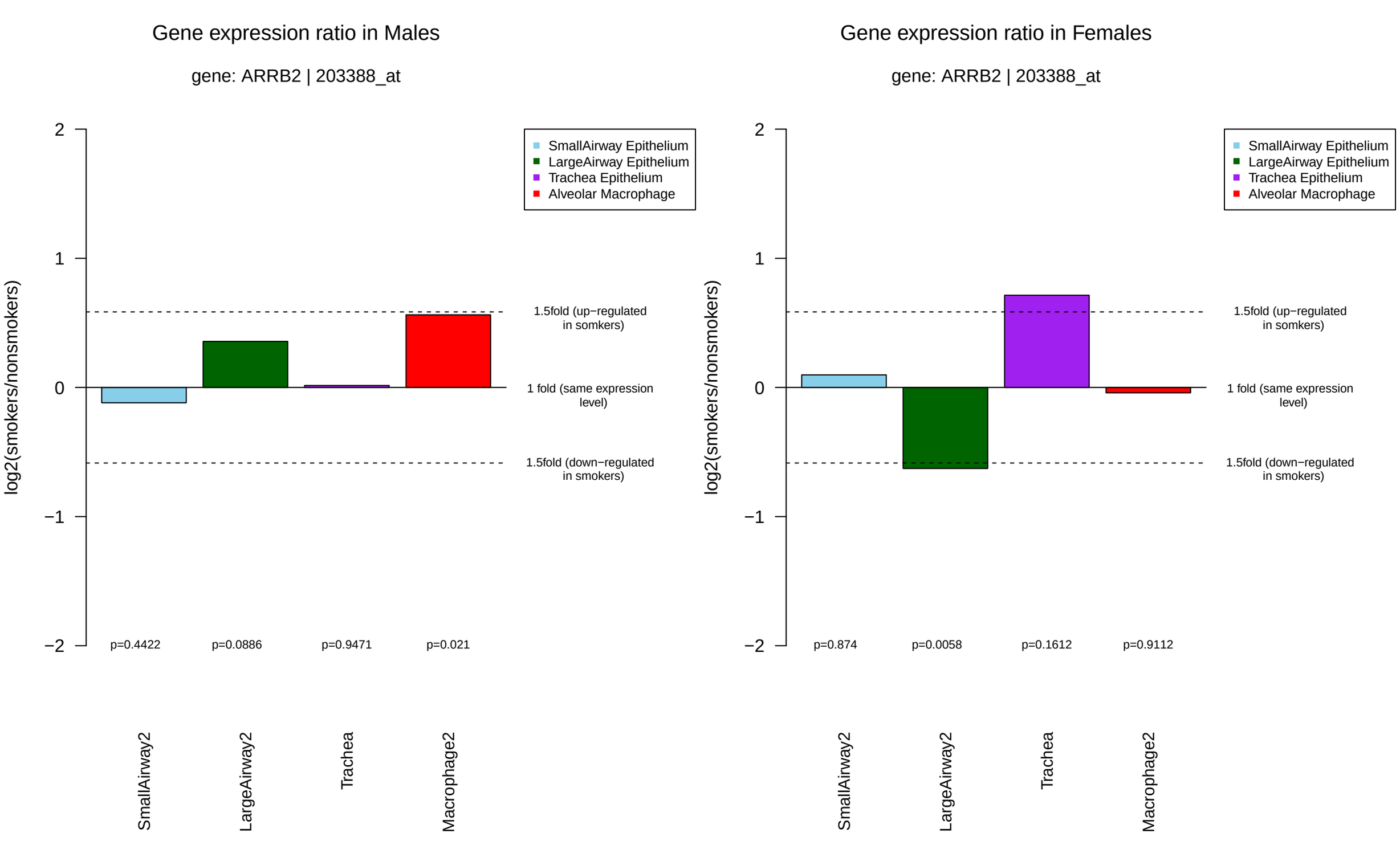
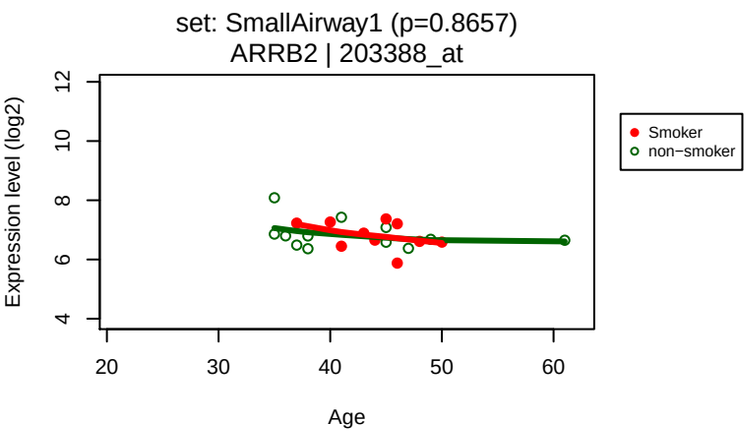
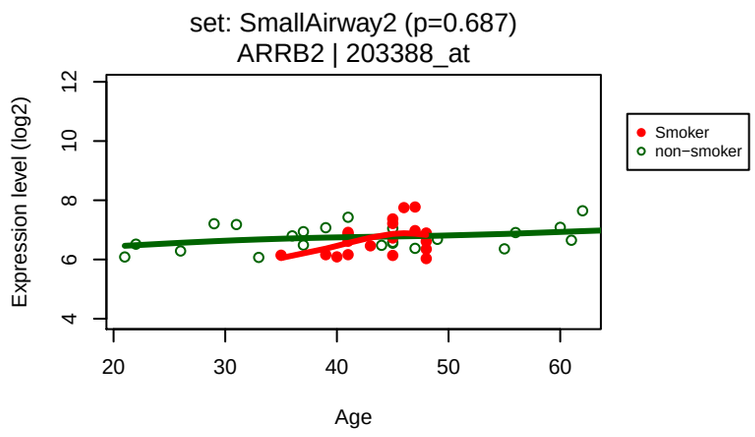
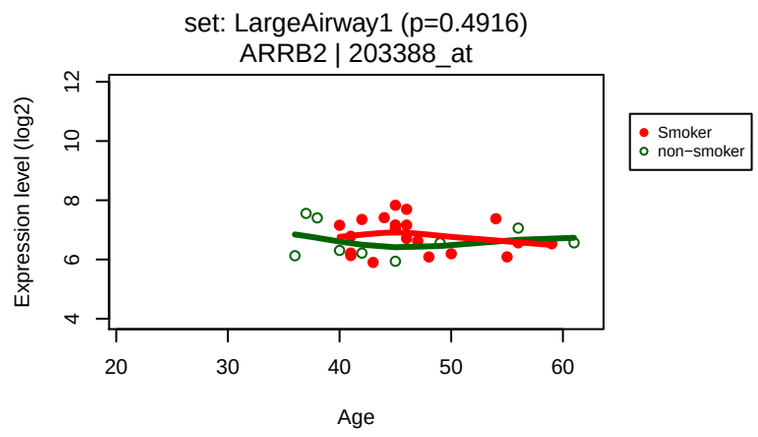

Gene expression of smokers and nonsmokers
gene: ARRB2 | 203388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8657
p=0.687
p=0.3914
p=0.4916
p=0.9168
p=0.5584
p=0.5764
p=0.0705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

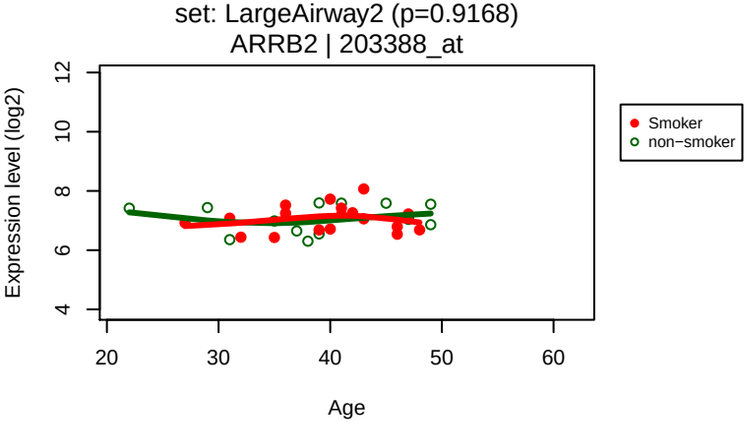

Gene expression ratio (smokers/nonsmokers)
gene: ARRB2 | 203388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8657
p=0.687
p=0.3914
p=0.4916
p=0.9168
p=0.5584
p=0.5764
p=0.0705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

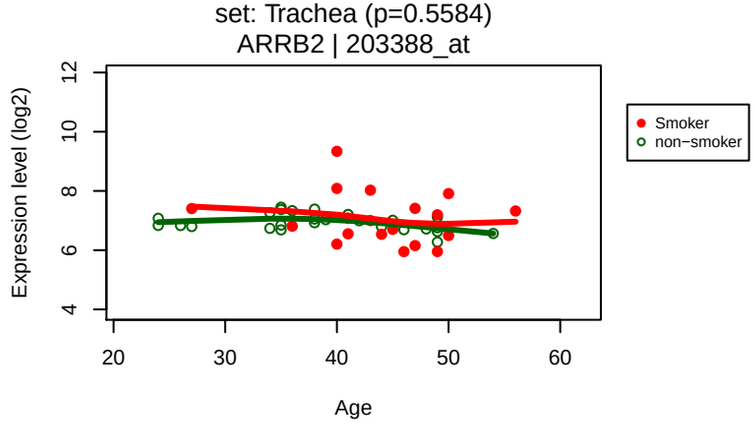

Gene expression ratio in Males
gene: ARRB2 | 203388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4422
p=0.0886
p=0.9471
p=0.021
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARRB2 | 203388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.874
p=0.0058
p=0.1612
p=0.9112
SmallAirway2
LargeAirway2
Trachea
Macrophage2

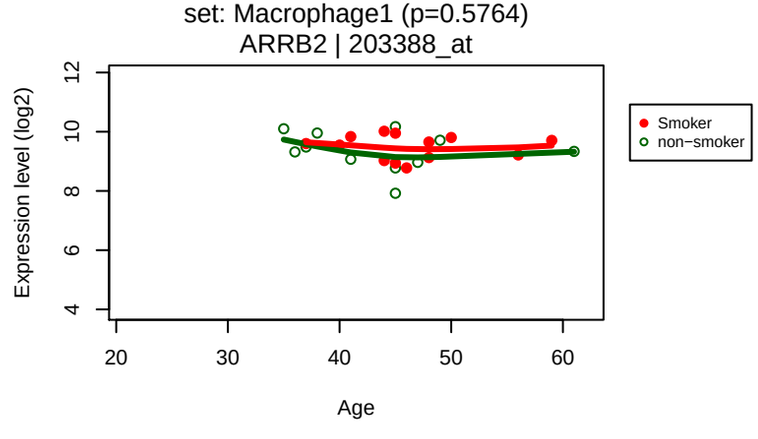

set: SmallAirway1 (p=0.8657)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker

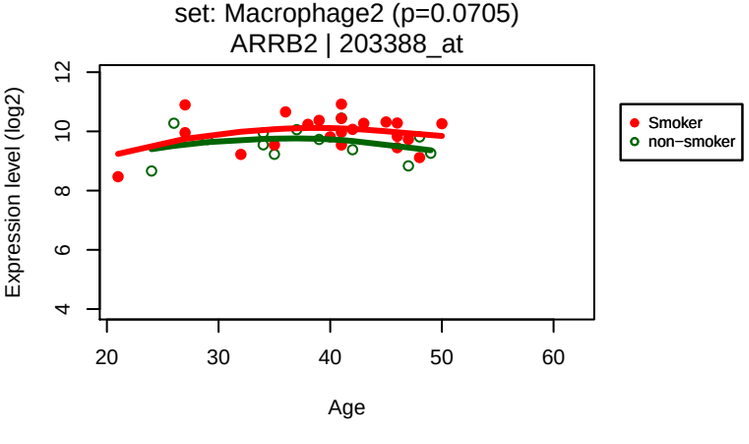

set: SmallAirway2 (p=0.687)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker

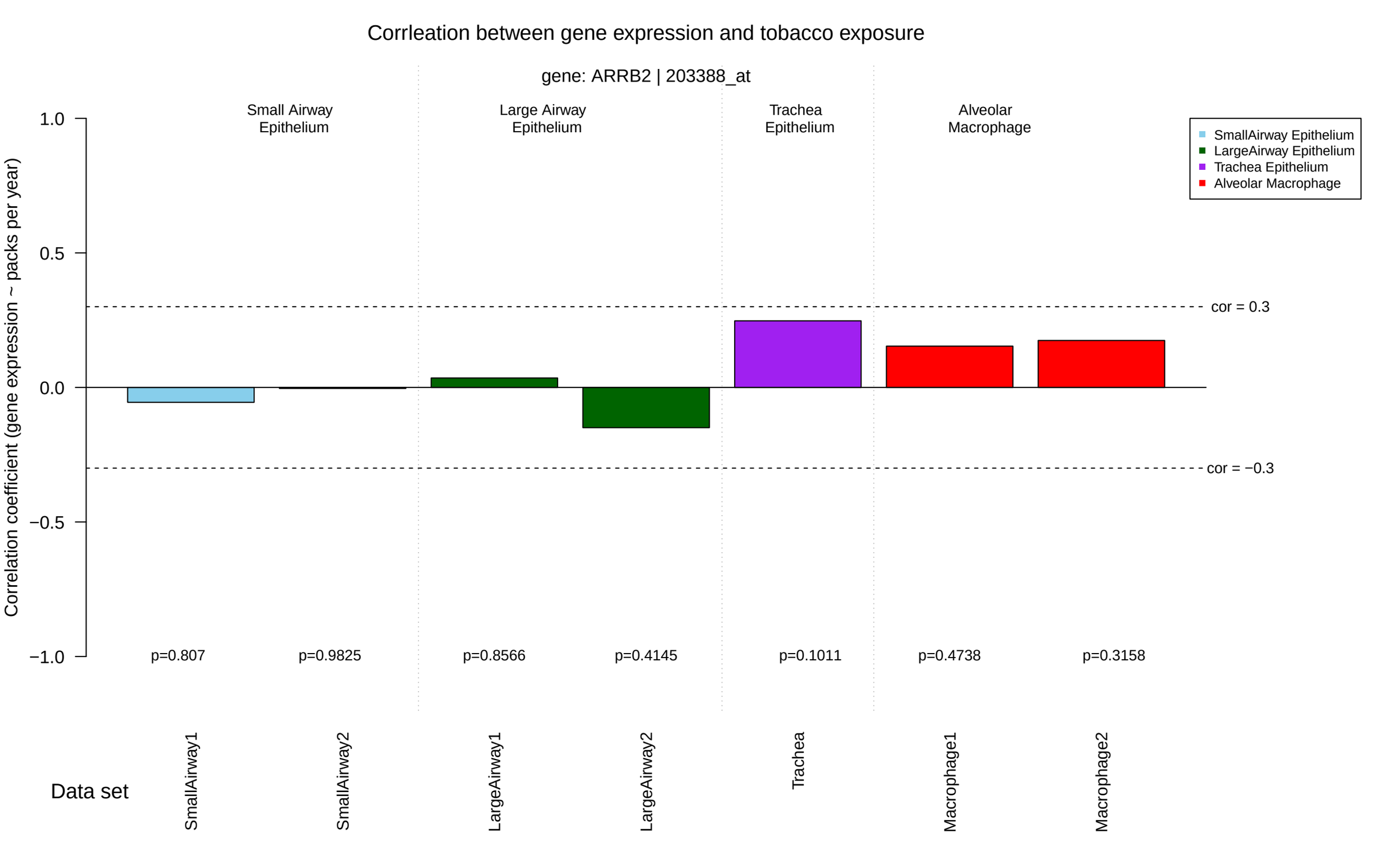

set: LargeAirway1 (p=0.4916)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9168)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5584)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5764)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0705)
ARRB2 | 203388_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARRB2 | 203388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.807
p=0.9825
p=0.8566
p=0.4145
p=0.1011
p=0.4738
p=0.3158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2